Letters

RESEARCH LETTER

HEALTH EQUITY

Public Opinion on Use of Race in Clinical Algorithms

The use of race in clinical algorithms remains debated. However, there are no large-scale quantitative data on patient preferences, an important consideration in algorithm design. Accordingly, we conducted a nationally representative survey of US adults to address this gap.

Methods | In September 2024, we invited adults (aged ≥18 years) to complete an online survey using YouGov, top-ranked polling platform.[1] Participants comprised a general US population sample and separate samples of self-reported Asian, Black, and Hispanic adults. Samples were weighted using propensity scores to account for sampling and nonresponse bias. Two-sided $P < .05$ was significant. Statistical analyses were performed with R, version 4.4.0 (R Project for Statistical Computing). The Cornell University Institutional Review Board deemed this deidentified survey research exempt from review; participants provided informed consent. The study adhered to the AAPOR reporting guideline. Survey questions and additional analytic details are provided in the eAppendix and eMethods in Supplement 1.

Results | Of 2638 invitees, 1997 (75.7%) completed the survey; 247 were excluded using attention and quality checks. Of the remaining 1750 respondents, the median (IQR) age was 45 (31-61) years; 916 (52.3%) were female and 834 (47.7%) were male (**Table**).

Most respondents were comfortable with the use of race in some circumstances. After reweighting, 79.8% (95% CI, 77.4%-82.1%) of adults preferred that their physician use all available information, including race, to make health decisions, whereas 20.1% (95% CI, 17.9%-22.6%) preferred that race be excluded (**Figure**, A). Similarly, 49.7% (95% CI, 46.8%-52.5%) of adults disagreed with the statement "There are no circumstances under

Table. Characteristics of Survey Respondents by Race and Ethnicity[a]

Characteristic	Unweighted						Overall weighted
	Asian (n = 275)	Black (n = 377)	Hispanic (n = 269)	White (n = 730)	Other (n = 99)[b]	Overall (N = 1750)	
Oversampling factor	2.16	1.73	1.37	0.63	1.73	1.00	NA
Completion time, median (IQR), min	9.5 (6.3-16.3)	13.1 (7.7-22.0)	10.5 (6.7-18.2)	12.0 (7.9-18.8)	11.4 (8.3-18.2)	11.6 (7.5-19.2)	11.7 (7.7-19.0)
Comprehension questions correct	233 (84.7)	298 (79.0)	221 (82.2)	660 (90.4)	89 (89.9)	1501 (85.8)	1501 (87.4)
Sex							
Female	155 (56.4)	183 (48.5)	146 (54.3)	374 (51.2)	58 (58.6)	916 (52.3)	916 (51.5)
Male	120 (43.6)	194 (51.5)	123 (45.7)	356 (48.8)	41 (41.4)	834 (47.7)	834 (48.5)
Age, y							
Median (IQR)	43 (31-57)	40 (27-59)	41 (29-56)	52.5 (35-64)	46 (34.5-61.5)	45 (31-61)	47 (32-63)
18-44	148 (53.8)	213 (56.5)	156 (58.0)	288 (39.5)	47 (47.5)	852 (48.7)	852 (46.1)
45-64	87 (31.6)	106 (28.1)	88 (32.7)	264 (36.2)	30 (30.3)	575 (32.9)	575 (32.6)
65-74	32 (11.6)	48 (12.7)	21 (7.8)	118 (16.2)	16 (16.2)	235 (13.4)	235 (14.9)
≥75	8 (2.9)	10 (2.7)	4 (1.5)	60 (8.2)	6 (6.1)	88 (5.0)	88 (6.4)
Educational attainment							
Less than high school	8 (2.9)	16 (4.2)	24 (8.9)	20 (2.7)	3 (3.0)	71 (4.1)	71 (6.8)
High school	267 (97.1)	361 (95.8)	245 (91.1)	710 (97.3)	96 (97.0)	1679 (95.9)	1679 (93.2)
4-y College	172 (62.5)	98 (26.0)	56 (20.8)	261 (35.8)	28 (28.3)	615 (35.1)	615 (34.3)
Family income, $							
0-79 999	112 (40.7)	272 (72.1)	190 (70.6)	400 (54.8)	79 (79.8)	1053 (60.2)	1053 (57.5)
≥80 000	121 (44.0)	65 (17.2)	53 (19.7)	258 (35.3)	10 (10.1)	507 (29.0)	507 (31.3)
Not reported	42 (15.3)	40 (10.6)	26 (9.7)	72 (9.9)	10 (10.1)	190 (10.9)	190 (11.2)
Employment status							
Full-time	119 (43.3)	119 (31.6)	112 (41.6)	291 (39.9)	33 (33.3)	674 (38.5)	674 (39.4)
Part-time	39 (14.2)	74 (19.6)	33 (12.3)	97 (13.3)	12 (12.1)	255 (14.6)	255 (12.9)
Not employed[c]	117 (42.5)	184 (48.8)	124 (46.1)	342 (46.8)	54 (54.5)	821 (46.9)	821 (47.7)

(continued)

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Table. Characteristics of Survey Respondents by Race and Ethnicity[a] (continued)

Characteristic	Unweighted						Overall weighted
	Asian (n = 275)	Black (n = 377)	Hispanic (n = 269)	White (n = 730)	Other (n = 99)[b]	Overall (N = 1750)	
Political party							
Democrat	104 (37.8)	201 (53.3)	105 (39.0)	229 (31.4)	36 (36.4)	675 (38.6)	675 (35.5)
Republican	49 (17.8)	37 (9.8)	61 (22.7)	252 (34.5)	17 (17.2)	416 (23.8)	416 (27.4)
Neither	122 (44.4)	139 (36.9)	103 (38.3)	249 (34.1)	46 (46.5)	659 (37.7)	659 (37.1)
Urbanicity							
Urban	80 (29.1)	189 (50.1)	106 (39.4)	195 (26.7)	33 (33.3)	603 (34.5)	603 (31.4)
Suburban	163 (59.3)	121 (32.1)	119 (44.2)	293 (40.1)	33 (33.3)	729 (41.7)	729 (41.2)
Rural	21 (7.6)	33 (8.8)	31 (11.5)	159 (21.8)	22 (22.2)	266 (15.2)	266 (18.0)
Not reported	11 (4.0)	34 (9.0)	13 (4.8)	83 (11.4)	11 (11.1)	152 (8.7)	152 (9.4)
Self-rated health							
Fair or poor	39 (14.2)	41 (10.9)	44 (16.4)	76 (10.4)	20 (20.2)	220 (12.6)	220 (12.1)
Good	72 (26.2)	79 (21.0)	68 (25.3)	148 (20.3)	23 (23.2)	390 (22.3)	390 (22.3)
Very good or excellent	120 (43.6)	116 (30.8)	86 (32.0)	195 (26.7)	26 (26.3)	543 (31.0)	543 (28.7)
Not reported	44 (16.0)	141 (37.4)	71 (26.4)	311 (42.6)	30 (30.3)	597 (34.1)	597 (36.9)
Health care experiences							
Told that their race has been used in medical decisions	30 (10.9)	67 (17.8)	30 (11.2)	40 (5.5)	10 (10.1)	177 (10.1)	177 (8.6)
Believe that their race has been used in medical decisions	40 (14.5)	93 (24.7)	40 (14.9)	108 (14.8)	16 (16.2)	297 (17.0)	297 (16.0)
Mostly or always asked about race in health care settings	99 (36.0)	140 (37.1)	94 (34.9)	286 (39.2)	41 (41.4)	660 (37.7)	660 (37.9)
Believe that having a physician of the same race or ethnicity affects quality of care	97 (35.3)	150 (39.8)	89 (33.1)	175 (24.0)	29 (29.3)	540 (30.9)	540 (27.9)

Abbreviation: NA, not applicable.

[a] Data are presented as No. (%) of participants unless indicated otherwise. Percentages are unweighted unless indicated otherwise.

[b] Includes American Indian or Alaska Native (n = 17), Middle Eastern (n = 3), multiracial (n = 54), and other race (n = 25).

[c] Includes student, homemaker, retired, disabled, unemployed, and other.

which a doctor should use my race in clinical care," whereas 22.0% (95% CI, 19.6%-24.4%) agreed (Figure, B). Preferences varied by race and ethnicity. For example, more Black (28.8% [95% CI, 23.7%-33.8%]) and Hispanic (30.4% [95% CI, 24.3%-36.5%]) respondents preferred that physicians exclude information on race compared with Asian (17.2% [95% CI, 12.1%-22.3%]) and White (17.1% [95% CI, 14.1%-20.1%]) respondents. When asked about a hypothetical cancer risk prediction algorithm, more respondents were uncomfortable with the use of income (39.3% [95% CI, 36.5%-42.1%]) and zip code (22.3% [95% CI, 19.9%-24.7%]) compared with race (8.8% [95% CI, 7.2%-10.5%]); fewer respondents were uncomfortable with the use of age or sex (Figure, C). Fewer respondents were uncomfortable with the use of race if physicians discussed how and why they were using it (8.3% [95% CI, 6.7%-9.9%]) vs using this information without asking (31.3% [95% CI, 28.6%-34.0%]) (Figure, D). However, few (8.6% [95% CI, 7.0%-10.2%]) reported ever being told that their race was used to make medical recommendations or decisions.

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Supplemental content

Discussion | Our nationally representative survey study had 4 key findings. First, most individuals preferred that physicians use all available information, including race, to inform health decisions. This finding highlights a tension between calls to eliminate uses of race in medicine and attitudes or understandings among the public, whose support may be needed for a lasting resolution. Second, discomfort with the use of race varied across groups, with Black and Hispanic respondents more often opposed than Asian and White respondents. Differing opinions may reflect lived experiences that vary by race, ethnicity, and other social factors.[2] Third, respondents were 4 times more likely to report discomfort when clinicians used race without asking. However, few respondents reported having ever been told that their race was used, despite earlier calls for transparency.[3] Fourth, respondents were less comfortable with using income or zip code than with using race, suggesting that replacement of race with other social determinants of health may not be seen as more trustworthy.[4]

This study has limitations. We only included US adults who use the internet (96%),[5] understand English well (92%),[6] and consented to participate in a standing survey panel. Responses to hypothetical questions may not align with preferences in clinical settings. Finally, patient preferences are not sufficient to justify the inclusion or exclusion of race and should be considered alongside other criteria, including accuracy, equity, and outcomes.

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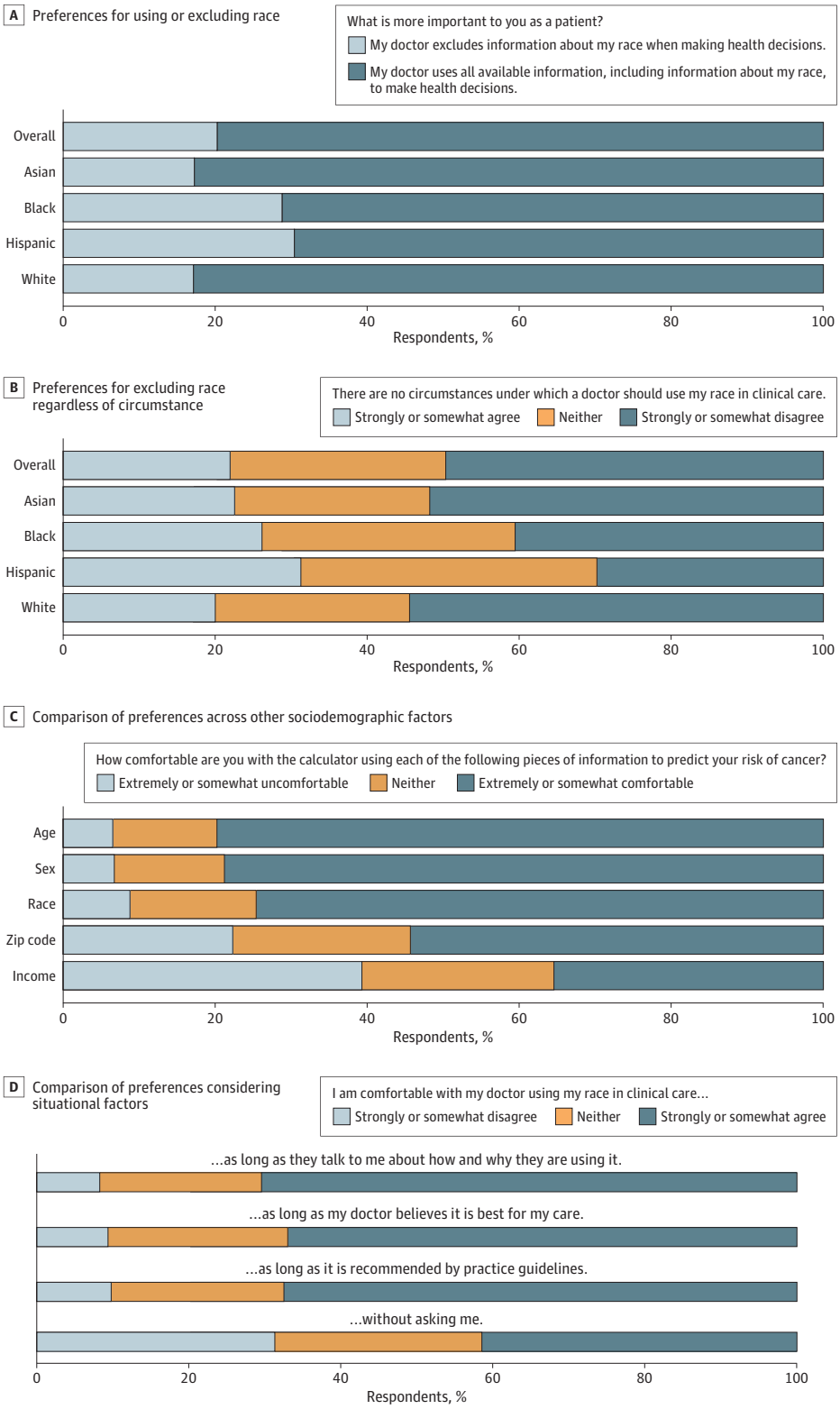
Figure. Survey Respondent Preferences on Use of Race in Clinical Algorithms



A, Preferences for using or excluding race. B, Preferences for excluding race regardless of circumstance. C, Comparison of preferences across other sociodemographic variables. D, Consideration of situational factors. Response percentages were calculated using survey weights to account for selection and nonresponse bias. Direct quotes from the survey are presented here; complete survey questions are provided in the eAppendix in Supplement 1.

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WOMEN'S HEALTH

Gestational Diabetes in the US From 2016 to 2024

Gestational diabetes (GD) affects the short- and long-term health of pregnant individuals and their child.[1] In the US, rates of GD increased substantially between 2011 and 2021 with variation by maternal race and ethnicity, but more recent data are not known.[2,3] To inform management and prevention strategies, we evaluated GD patterns for race and ethnicity groups between 2016 and 2024, including during and after the COVID-19 pandemic.

Methods | We included data on all US residents aged 15 to 44 years with a singleton first live birth between January 1, 2016 (first year all states and regions used the 2003 birth certificate revision), and December 31, 2024 (provisional data at the time of analysis, with >99.9% of all 2024 birth records received and processed), from the National Center for Health Statistics birth certificates.[4] We excluded individuals with missing data on GD status. GD information on a birth certificate is recommended to be completed by the professional birth attendant based on the mother's medical records indicating a diagnosis of glucose intolerance requiring treatment during the pregnancy. The Northwestern University Institutional Review Board deemed this cross-sectional study exempt from review and informed consent because public, deidentified data were used. We followed the STROBE reporting guideline.

Supplemental content

We calculated age-standardized GD rates per 1000 live births (standardized to the age distribution of singleton births with GD data in 2016). Age-standardized rates were calculated for the overall population, by 6 primary racial and ethnic groups (Hispanic, non-Hispanic American Indian or Alaska Native [hereafter American Indian or Alaska Native], non-Hispanic Asian [hereafter Asian], non-Hispanic Black [hereafter Black], non-Hispanic Native Hawaiian or Other Pacific Islander [hereafter Native Hawaiian or Other Pacific Islander], and non-Hispanic White [hereafter White]), by individual ethnicities in the Hispanic group (Central and South American, Cuban, Dominican, Mexican, and Puerto Rican), and by individual ethnicities in the Asian group (Asian Indian, Chinese, Filipina, Japanese, Korean, and Vietnamese) because prior research demonstrated heterogeneity in GD rates and outcomes by race and ethnicity.[3] Race and ethnicity were self-reported by selecting from fixed categories.

Joinpoint Regression 5.4.0 (National Cancer Institute) was used to estimate the average annual percentage change (AAPC) of GD rates from 2016 to 2024 for these racial and ethnic groups. Two-sided $P < .05$ indicated AAPC values significantly different than 0 calculated via the empirical quantile method. Statistical analysis was conducted in August 2025.

Results | Among 12 617 106 mothers (mean [SD] age at delivery, 27.1 [5.8] years) included, 0.6% identified as American

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